Low-pass and High-pass Filtering of Sentinel-2 Imagery over the Aravalli Ridge

Which filtering technique suits which remote sensing application?

Group 1

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EES403: Advanced Remote Sensing and GIS
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1. Introduction — spatial frequency, aim
2. Study Area and Data — scene, reference map
3. Methods — filters, experiments
4. Results I — images, ringing, E1
5. Results II — E2 to E5
6. Discussion and Conclusions — decision matrix

Presenter

Sangeeta
Sangam
Pranay
Aparna
Robin
Sangeeta

The talk follows the report: Chapters 1 to 6, one section each.

Section 1: Introduction

Presented by **Sangeeta**

1. Introduction
2. Study Area and Data
3. Methods
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6. Discussion and Conclusions

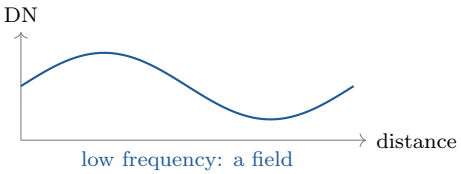
Spatial frequency is how often brightness changes per unit distance

Low spatial frequency: grey values change gradually
fields, water, broad slopes

High spatial frequency: grey values change abruptly
roads, roofs, ridge crests, field boundaries, noise

Every image = broad pattern (low) + edges and texture (high).
A filter keeps one part and suppresses the other.

Definition after Lillesand et al. (2015).



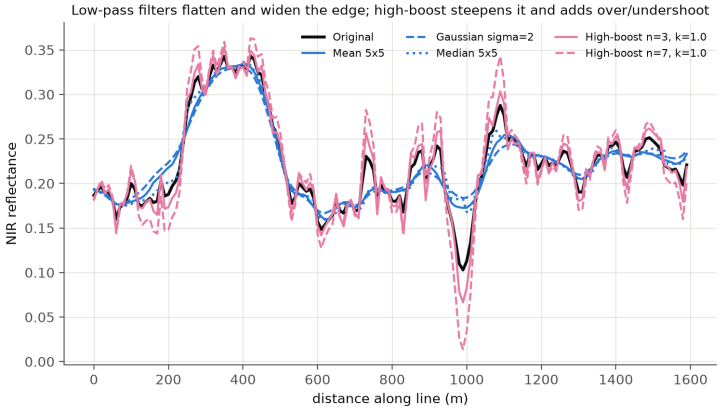


Low-pass flattens edges; high-pass and high-boost sharpen them

Low-pass (LPF)
averages neighbours
⇒ smoothing, less noise

High-pass (HPF)
HPF = original − LPF
⇒ only edges remain

High-boost
original + k·HPF
⇒ sharper, colours kept



NIR reflectance along a 1.6 km line across a cleared patch on the ridge (report Fig. 1.1).

Both can be computed in the *spatial* domain (kernels) or the *frequency* domain (FFT masks).

Aim: decide which filter is most suitable, application by application

Smoothing and edge enhancement do opposite jobs, so we split the brief into five questions:

Experiment	Question
E1	Which filter removes noise best?
E2	Which filter helps automatic land-cover classification most?
E3	Which filter makes edges, boundaries and lineaments clearest?
E4	How much does each filter change the reflectance values?
E5	Is the spatial or the frequency route faster?

The answers are combined in a **decision matrix**: one best filter per application.

Section 2: Study Area and Data

Presented by **Sangam**

1. Introduction

2. Study Area and Data

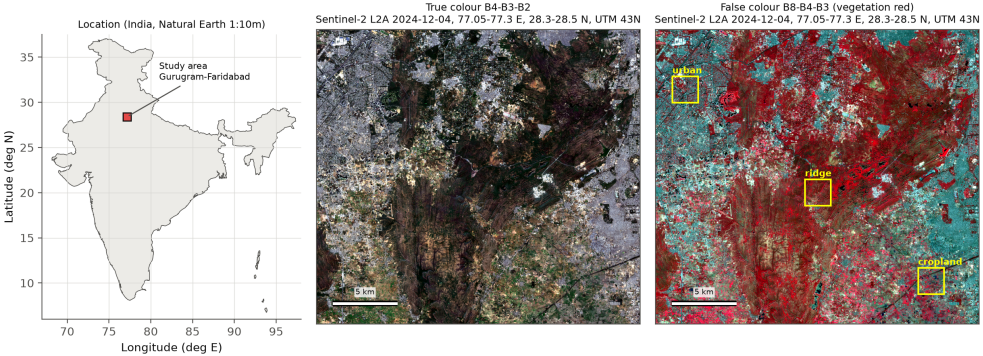
3. Methods

4. Results I: Filtered Images, Ringing and Noise

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The Aravalli ridge holds ridges, city and farmland in one scene

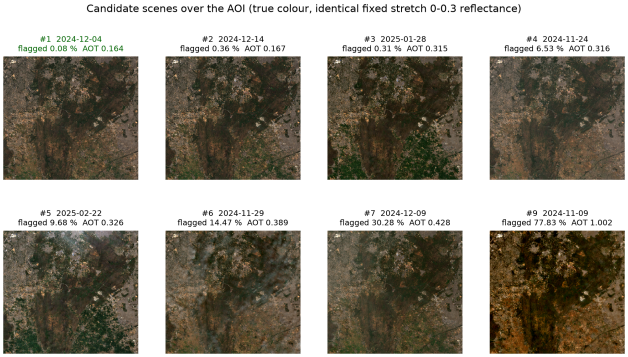


Left: location. Centre: true colour. Right: false colour (NIR, red, green), vegetation red. Yellow boxes: ridge, urban and cropland windows.

Area: 77.05–77.30°E, 28.30–28.50°N, 25 × 23 km
Low frequency: crop parcels in the south-east

High frequency: folded quartzite ridges, Gurugram streets
Also: old quarries, a few ponds

Scenes were ranked by cloud and haze inside the study area

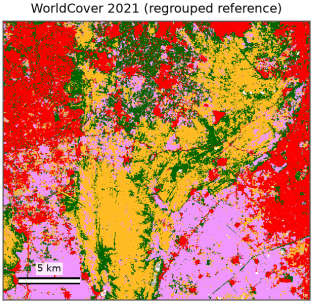


Candidate scenes over the study area, same colour stretch; flagged share and mean AOT above each.

- Sentinel-2 Level-2A (Drusch et al., 2012), Google Earth Engine
- Nov 2024 – Feb 2025, scene cloud $< 5\%$: 11 scenes
- Flagged pixels *inside the box*: 0.08% to 97.2%
- Ranked by scene classification, s2cloudless and aerosol optical thickness (AOT)

Chosen scene	2024-12-04, tile 43RGM
Flagged pixels	0.08%, AOT 0.164
Bands	B2, B3, B4, B8 at 10 m
Grid	2491 × 2261 px, UTM 43N
Reflectance	DN / 10000, clipped to [0, 1]

ESA WorldCover 2021 provides the land-cover reference



WorldCover v200 (Zanaga et al., 2022), 10 m, regridded to the Sentinel-2 grid (nearest neighbour), classes merged:

Class	Share of area
Trees	19.9%
Shrub/Grass	28.7%
Cropland	24.6%
Built-up	25.3%
Bare/Mining	1.2%
Water (dropped, $< 0.5\%$)	0.19%

WorldCover is itself a classifier output: a *reference map*, not ground truth.

Section 3: Methods

Presented by **Pranay**

1. Introduction

2. Study Area and Data

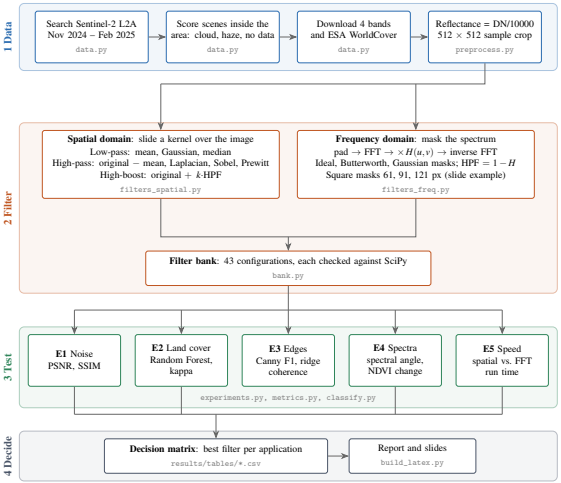
3. Methods

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Workflow: data, filter, test, decide



- Python package `rsfilter`; grey text in each box names the module
- All filters written in NumPy, per band, on float32 reflectance
- Unit tests against `scipy.ndimage`: max. difference $< 10^{-5}$
- Mirror (reflect) padding at the image border
- One command, `python scripts/run_all.py`, repeats everything; seed 42

Spatial domain: each output pixel is a weighted sum of its neighbours

$$g(x,y) \;=\; \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\; f(x-s,\; y-t)$$

- f: input band, w: kernel of size $(2a+1) \times (2b+1)$, g: output
- Convolution flips the kernel by 180°; irrelevant for our symmetric kernels
- **Shift and accumulate**: pad once, then for every kernel cell (s,t)
  $g \mathrel{+}= w(s,t)\cdot$ (padded image shifted by (s,t))
  $\Rightarrow$ loop over $(2a+1)(2b+1)$ cells, not over millions of pixels
- Cost per pixel $\propto$ number of kernel cells $= n^2$ for an $n \times n$ kernel

Gonzalez and Woods (2018), Ch. 3.

Spatial kernels: low-pass averages, high-pass differences

Low-pass

Mean: $w_{\text{mean},3} = \frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$, $n = 3, 5, 7, 9$

Gaussian: $w(s,t) \propto \exp\!\left(-\frac{s^2+t^2}{2\sigma^2}\right)$
side $2\lceil 3\sigma \rceil + 1$, $\sum w = 1$, $\sigma = 1, 2, 3$

Median ($n = 3, 5$): median of the window; non-linear, ignores single extreme values

High-pass

$\text{HPF}_n = f - \text{mean}_{n\times n}(f)$ (Lillesand et al., 2015)

$$w_{\text{Lap},4} = \begin{bmatrix}0 & 1 & 0\\ 1 & -4 & 1\\ 0 & 1 & 0\end{bmatrix} \quad w_{\text{Sobel},x} = \begin{bmatrix}-1 & 0 & 1\\ -2 & 0 & 2\\ -1 & 0 & 1\end{bmatrix}$$

Gradient magnitude $\sqrt{G_x^2 + G_y^2}$ (G_y: transposed kernel); Prewitt uses weights 1, 1, 1

High-boost: enhanced $= f + k \cdot \text{HPF}_n$, $k = 0.5, 1, 1.5$

Frequency domain: filtering becomes a multiplication of the spectrum

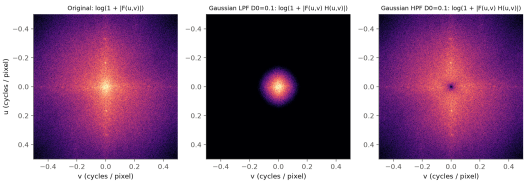
$$g = \mathcal{F}^{-1}\big\{H(u,v) \cdot \mathcal{F}\{f\}\big\}$$

Pipeline per band: mirror-pad 64 px → FFT → shift centre → $\times H$ → shift back → inverse FFT → real part → crop.

D: distance from spectrum centre, $D = 1$ at Nyquist; cutoff $D_0 = 0.1, 0.2, 0.3$

Ideal:	$H = 1$ if $D \le D_0$, else 0
Butterworth ($n = 2$):	$H = 1/\big(1 + (D/D_0)^4\big)$
Gaussian:	$H = \exp\big(-D^2/2D_0^2\big)$
High-pass:	$H_{\mathrm{HP}} = 1 - H$

Gaussian mask ≡ spatial Gaussian with $\sigma = 1/(\pi D_0)$ px



Log-magnitude spectra of the NIR crop: original, Gaussian LPF and HPF ($D_0 = 0.1$).

A sharp cut in H ⇒ a sinc-like kernel with negative side lobes ⇒ ripples (*ringing*, Gibbs, ≈ 9% overshoot).

Filter bank: 43 configurations compared

Family	Filter	Parameters
Low-pass, spatial	Mean; Gaussian; Median	$n = 3, 5, 7, 9$; $\sigma = 1, 2, 3$; $n = 3, 5$
Low-pass, frequency	Ideal, Butterworth, Gaussian	$D_0 = 0.1, 0.2, 0.3$
High-pass, spatial	Original $-$ mean; Laplacian; Sobel, Prewitt	$n = 3, 5, 7$; 4-, 8-neighbour
High-pass, frequency	Ideal, Butterworth, Gaussian	$1 - H$, $D_0 = 0.1, 0.2, 0.3$
High-boost	$f + k \cdot \mathrm{HPF}_n$	$n = 3, 5, 7$; $k = 0.5, 1.0, 1.5$

Plus the slide example: square ideal low-pass masks of 61, 91 and 121 px on a 512×512 crop.

Five experiments, each with a measure that decides the winner

	Set-up	Measure
E1	Gaussian noise $\sigma = 0.02$ or 2% salt-and-pepper on a clean 512×512 crop; every LPF	PSNR $= 10 \log_{10}(1/\mathrm{MSE})$ dB; SSIM (Wang et al., 2004)
E2	Random Forest (Breiman, 2001), 200 trees, 2000 px/class; 4 bands + NDVI after filtering	$\kappa = (p_o - p_e)/(1 - p_e)$; accuracy; macro-F1; salt-and-pepper index
E3	NIR band	mean gradient; Canny (Canny, 1986) edge F1 vs WorldCover boundaries (2 px); ridge coherence
E4	all bands	SAM $= \arccos \frac{\mathbf{x}\cdot\mathbf{y}}{\lvert\mathbf{x}\rvert\lvert\mathbf{y}\rvert}$; $\lvert\Delta\mathrm{NDVI}\rvert$; RMSE
E5	one full band, 2261×2491 px	median of 5 runs, spatial vs FFT

Two design choices that keep the evaluation honest

E2: spatially blocked split

Neighbouring pixels are near-copies, so a random pixel split inflates accuracy.

$\Rightarrow$ 128 $\times$ 128 px blocks, 70% train / 30% test (10 000 training, 1 700 310 test pixels)

E3: orientation coherence

Structure tensor $J = G_\rho * \begin{bmatrix} g_x^2 & g_x g_y \\ g_x g_y & g_y^2 \end{bmatrix}$,

eigenvalues $\lambda_1 \geq \lambda_2$

$$C = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2} \in [0, 1]$$

1: parallel ridges; 0: speckle. 2 $\times$ 2 km ridge window (Bigün and Granlund, 1987).

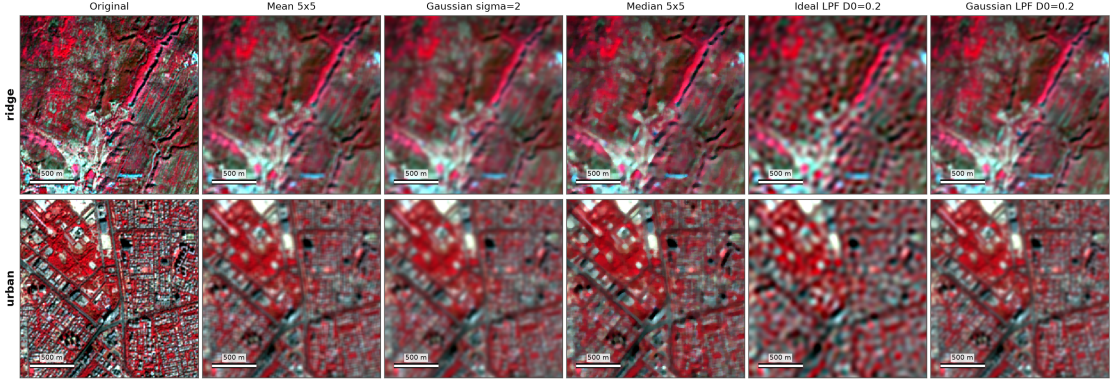
No published lineament map exists here, so coherence stands in for lineament quality.

Section 4: Results I: Filtered Images, Ringing and Noise

Presented by **Aparna**

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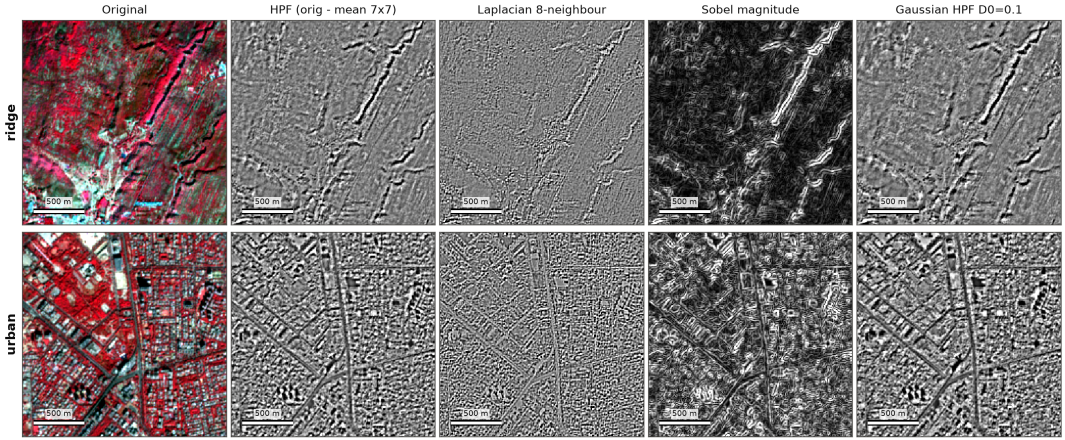
Low-pass filters keep the broad ridge but blur streets and field edges



2 km windows, ridge (top) and Gurugram (bottom); false colour, same stretch in every panel.

The median keeps straight edges sharper than a mean or Gaussian of the same size.

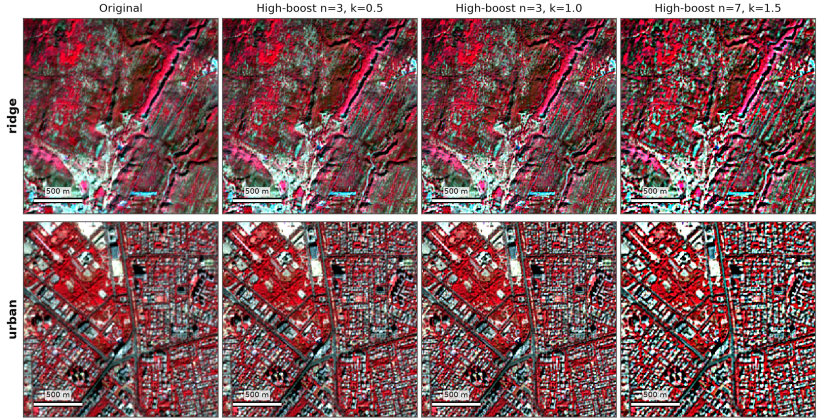
High-pass filters turn flat ground grey and keep only the edges



NIR band, same two windows.

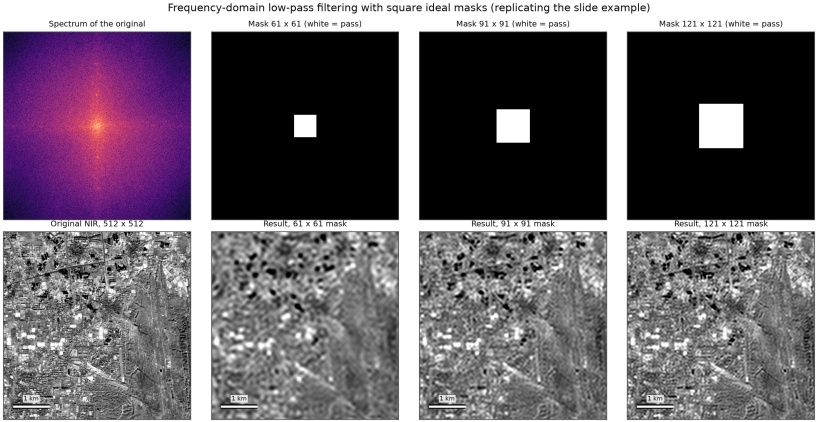
7×7 HPF and Sobel trace the ridge crests; the Laplacian reacts mostly to fine speckle.

High-boost sharpens boundaries and keeps the colours



At $k = 1.5$, bright and dark halos appear along roads and ridge crests.

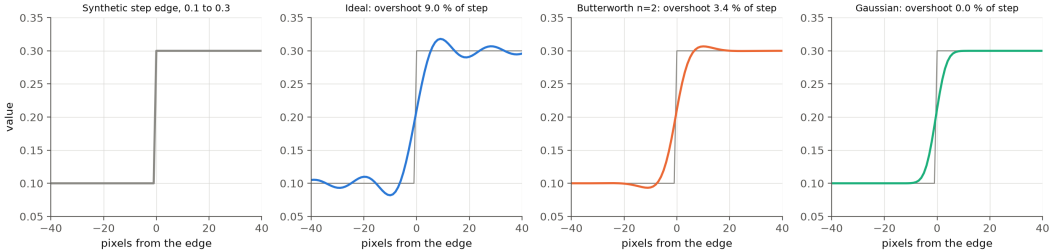
The slide example reproduced: smaller square mask, blurrier image



Square ideal low-pass masks of 61, 91 and 121 px (top) and the filtered NIR crop (bottom).

All three results show a faint ripple texture.

A sharp frequency cut causes ringing; smooth masks do not



The same mask code applied to a clean synthetic step edge, $D_0 = 0.1$.

Mask ($D_0 = 0.1$)	Ideal	Butterworth	Gaussian
Overshoot, % of step height	**9.0**	3.4	0.0

Use Gaussian or Butterworth masks, never the ideal mask.

E1: the best noise filter depends on the type of noise

	Gaussian, $\sigma = 0.02$		Salt-and-pepper, 2%	
Filter	PSNR (dB)	SSIM	PSNR (dB)	SSIM
Noisy input (no filter)	33.98	0.825	21.01	0.603
Mean 3×3	35.52	0.882	29.29	0.714
Mean 5×5	32.60	0.775	30.39	0.695
Gaussian $\sigma = 1$	**35.56**	**0.885**	30.27	0.748
Gaussian $\sigma = 2$	32.26	0.765	30.91	0.728
Median 3×3	35.19	0.872	**36.49**	**0.910**
Median 5×5	32.65	0.779	32.89	0.794
Ideal LPF $D_0 = 0.3$	33.05	0.799	29.65	0.676
Butterworth LPF $D_0 = 0.3$	34.33	0.848	30.71	0.746
Gaussian LPF $D_0 = 0.3$	35.22	0.876	30.48	0.749

Random noise: only light filters help; 13 of 18 made it worse.

Impulse noise: median 3×3 raises PSNR from 21.01 to 36.49 dB, 5.42 dB ahead of the best linear filter.

Median 3×3 for impulse noise; Gaussian $\sigma = 1$ or mean 3×3 for random noise.

Section 5: Results II: Classification, Edges and Cost

Presented by **Robin**

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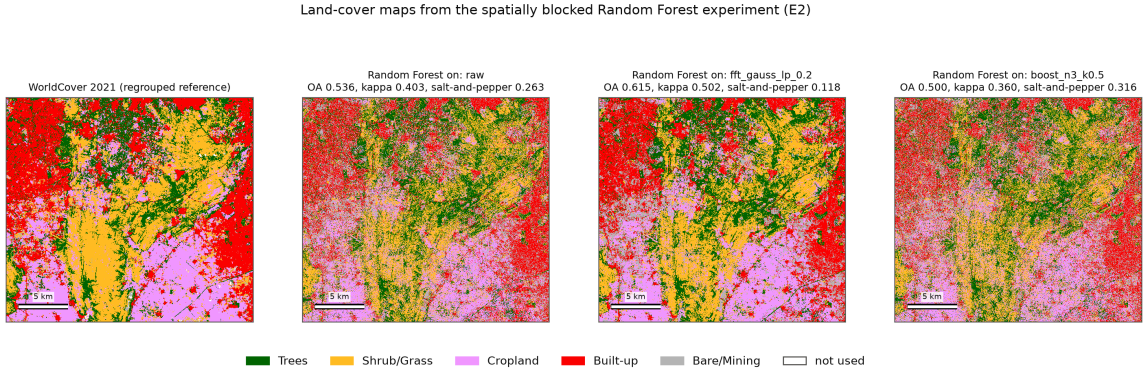
E2: low-pass inputs classify best, high-boost inputs worst

Input	Family	Accuracy	κ	Macro-F1	Salt-and-pepper
No filter	none	0.536	0.403	0.473	0.263
Gaussian $\sigma = 2$	LPF, spatial	0.613	0.500	0.539	**0.113**
Gaussian LPF $D_0 = 0.2$	LPF, frequency	**0.615**	**0.502**	**0.540**	0.118
High-boost $n = 3$, $k = 0.5$	high-boost	0.500	0.360	0.444	0.316
High-boost $n = 7$, $k = 1.5$	high-boost	0.431	0.275	0.383	0.378
Raw + HPF 7×7 layers	raw + HPF stack	0.598	0.481	0.528	0.178

Held-out blocks. Best and worst of each family; bold = best (lowest for salt-and-pepper).

Smoothing removes within-class variation (roof types, shadows, mixed pixels) before classification.

E2: smoothing also halves the speckle in the classified map



Salt-and-pepper index: raw 0.263 → best low-pass 0.118.

E3: high-pass helps boundaries and ridge lineaments

Boundaries (Canny edge F1)

No filter	0.675
High-boost $n = 5$, $k = 1.5$	**0.742**
Gaussian $\sigma = 1$	0.479
Mean 5×5	0.247
Built-up only: no filter	0.543
Built-up only: best	0.545

Ridge lineaments (coherence)

No filter	0.510
Prewitt	**0.536**
Sobel	0.535
HPF 7×7	0.502
Laplacian (4-nb.)	0.426
Gaussian $\sigma = 3$	0.754*

*high only because smoothing erases speckle

Smoothing removes edges; high-pass filtering helps wherever the task is *where things change*.

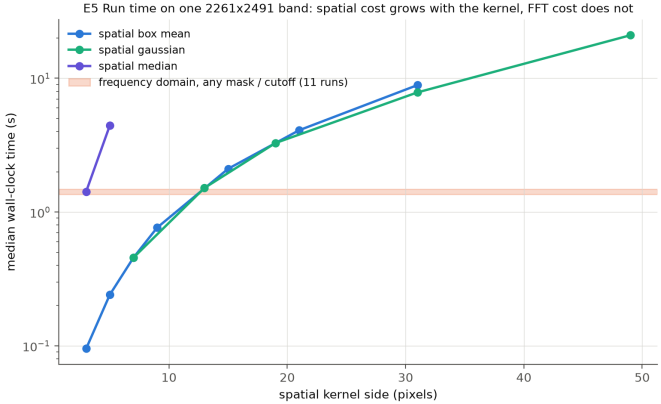
E4: light filters keep the spectra; HPF layers are not reflectance

Filter	Spectral angle (°)	$\lvert\Delta\text{NDVI}\rvert$
High-boost $n = 3$, $k = 0.5$	1.20	–
Median 3×3	2.22	0.033
Gaussian $\sigma = 1$	2.30	0.037
Sobel	18.90	–
HPF 7×7	88.76	–

Mean spectral angle to the original image; 0° = spectrum shape unchanged.

Keep high-pass products as *extra layers* beside the bands, never in place of them.

E5: FFT cost is flat, convolution cost grows with kernel size



One full band, 2261×2491 px:

- Spatial mean 3×3: 0.10 s
- Spatial mean 31×31: 8.88 s
- Spatial Gaussian 49×49: 20.96 s
- Any FFT mask: 1.36–1.48 s

Break-even near a 13-px kernel

Small kernels: convolve. Large kernels or many cutoffs: use the FFT.

Section 6: Discussion and Conclusions

Presented by **Sangeeta**

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Decision matrix: one best filter per application

Application	Best filter	Evidence (best vs. no filter)
Noise removal, random noise	Gaussian $\sigma = 1$	PSNR (dB): **35.56** vs 33.98
Noise removal, impulse noise	Median 3×3	PSNR (dB): **36.49** vs 21.01
Land-cover classification	Gaussian LPF $D_0 = 0.2$	Cohen's κ: **0.502** vs 0.403
Lineament mapping (ridges)	Prewitt	ridge coherence: **0.536** vs 0.510
Urban boundary delineation	High-boost $n = 3$, $k = 0.5$	built-up edge F1: **0.545** vs 0.543
Visual interpretation	High-boost $n = 3$, $k = 1.0$	NIR mean gradient: **0.0417** vs 0.0245

Generated by the analysis code from the experiment tables (`decision_matrix.csv`).

Low-pass when pixel values feed a calculation; **high-pass** when edges and structure are the point.

Recommendation

Quantitative analysis

classification, indices, change detection

$\Rightarrow$ light **Gaussian**, $\sigma \approx 1.5$–2 px (or $D_0 \approx 0.2$)
$\Rightarrow$ **median** 3×3 for impulse noise

Interpretation and mapping

visual analysis, boundaries, lineaments

$\Rightarrow$ **high-boost**, $k = 0.5$–1, for display
$\Rightarrow$ **Sobel / Prewitt** layer for lineaments

Avoid

the ideal frequency mask (ringing), and heavy smoothing (detail is lost before noise)

Limitations and conclusions

Limitations

- WorldCover is a model output, not ground truth
- One winter scene, one area
- One block split: κ differences ≈ 0.01 are not a ranking
- Coherence is a proxy for lineament quality
- Bare/Mining (1.2%): F1 ≤ 0.14 for every input

Conclusions

- No single filter is "most suitable"; the two families do opposite jobs
- Smoothing: κ +0.099, half the map speckle
- Sharpening: best boundary match and sharpest image
- Spatial and FFT routes agree for smooth masks

Thank you. Questions?

Live demo: `python -m rsfilter demo` (offline, 512×512 sample)

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